ARTCRAFT FLEET TECHNICAL BROCHURE · SERIES #23

AssetCraft v0.1.1

Fixed assets with Austrian AfA, software licences and expiry dates for a small business. One SQLite file, one binary that is both CLI and MCP server. Part of the back-office craft suite (replaces the asset Excel, the licence spreadsheet and the calendar reminders for certificates and Pickerl).



v0.1.1	3 Crates	3688	6-Gate
RELEASE VERSION	WORKSPACE TIERS	PURE-RUST LOC	ARTCRAFT VERIFIED

Repository: `D:\Dev\repos\assetcraft` · **Standard:** `CRAFT_BROCHURE_STANDARD.md` · **Compiled via:** `BookCraft v0.5.0`

Fixed assets with Austrian AfA, software licences and expiry dates for a small business. One SQLite file, one binary that is both CLI and MCP server. Part of the back-office craft suite (replaces the asset Excel, the licence spreadsheet and the calendar reminders for certificates and Pickerl).

Status: **v0.1.1, Tier 1** (store + CLI + MCP, tested). No desktop window yet. Synthetic demo data only; AfA rules not reviewed by a Steuerberater.

- How to use it day to day: HELP.md
- What it is for and where it stops: docs/PRD.md
- Every MCP tool: docs/TOOLS.md (generated, checked by a test)
- Changes: CHANGELOG.md

What it does

- **Assets:** tag, category, in-service date, cost (the AfA base), holder, location, serial, warranty, supplier, invoice reference, maintenance log.
- **AfA (EStG section 7), in cents:** straight-line, calendar fiscal year, Halbjahres-AfA in the first and in the disposal year, GWG write-off at or below the limit (default 1.000,00 since 2023, configurable). The remainder cent goes to the final year, so a schedule always sums to the cost and never grows a stray year.
- **Categories:** default life and minimum life (cars: at least 8 years, EStG section 8(6)) and the EKR asset account (IT 0620, car 0630, BGA 0660, GWG 0680). Editable.
- **buchcraft link:** `afa 2026 --emit` sends one `asset.depreciation_due` per asset to buchcraft (7000 or 7010 against the asset account). buchcraft turns them into drafts a human approves. Each asset/year is sent once; after that the AfA basis is frozen (in code and by a SQLite trigger).
- **Disposal:** disposal-year AfA, book value, gain or loss; the asset becomes read-only.
- **Licences:** seats, assignments (refused when full unless forced), renewal dates, cost per used seat. Keys are never stored, only where they live; a pasted key is refused.
- **Expiries:** warranties, licence renewals and deadlines (certificates, domains, insurance, inspections) due within N days, overdue first.
- **Labels:** printable 60 x 25 mm SVG with a QR code (`ASSET:<tag>`).
- **CSV import:** `tag;name;category;in_service;cost[;serial;location;holder]`, all-or-nothing, with `--dry-run`. Gets a real inventory in on day one.

Quick start

```
assetcraft demo-seed
assetcraft list
assetcraft schedule NAS-001
assetcraft afa 2026                     # preview
assetcraft afa 2026 --emit              # drafts for buchcraft
buchcraft inbox-drain; buchcraft drafts list; buchcraft drafts approve 3
assetcraft expiries --days 60
assetcraft license audit
assetcraft add GOL-002 "Mini PC" --category it --in-service 09.11.2026 --cost "689,00"
assetcraft label GOL-001 D:\Temp\gol-001.svg
assetcraft import D:\Temp\inventory.csv --dry-run
```

Database: `--db PATH`, else `$ASSETCRAFT_DB`, else `%LOCALAPPDATA%\assetcraft\assetcraft.db`.

MCP

```
{ "mcpServers": { "assetcraft": { "command": "D:/Dev/cargo-target/release/assetcraft.exe",
"args": ["mcp", "--allow-path", "D:/Dev/assetcraft-files"] } } }
```

24 tools, all `asset_*`; see docs/TOOLS.md.

Build

```
just ci
just build-release
```

Needs `../craft-biz` checked out next to this repo.

Not done (honest list)

- No egui desktop app.
- Degressive AfA (section 7(1a)), Luxustangente for cars (40.000 EUR cap), investment allowance (IFB) and non-calendar fiscal years are not modelled.
- Disposal sends no event: book the residual value (Anlagenabgang) and proceeds in buchcraft yourself.
- No direct import from rust-devices-mcp or winops inventory yet; export those to CSV and use `import`.
- Label is SVG only (print from a browser); no label-printer driver.

Product Requirements & Architecture Specification

What it is

Fixed assets with Austrian AfA, software licences and expiry dates for a small business. Pure Rust, ArtCraft pattern. Suite plan: `BACKOFFICE_CRAFT_SUITE.md`, section 4.4.

Users

- The owner who has to know what the business owns, what it is worth and what expires.
- Agents over MCP: register purchases, run the year-end AfA, watch expiries and licence seats.
- buchcraft (via the bus) and, through it, the Steuerberater.

Jobs

1. Register assets once and get a correct AfA plan.
2. Post year-end AfA to the books without retyping.
3. Not overpay or under-license software.
4. Never miss a warranty, certificate, domain, insurance or inspection date.

Today (v0.1.0)

- Assets with categories (default and minimum life, EKR account), maintenance log, disposal with gain/loss.
- AfA engine in cents: Halbjahres-AfA, GWG with configurable limit, remainder to the final year.
- `afa --emit` -> buchcraft drafts -> approved -> booked (verified end to end); posted AfA basis frozen by trigger.
- Licences with seats, assignment guard, audit; keys never stored.
- Expiries across warranties, renewals and deadlines; QR SVG labels; all-or-nothing CSV import.
- 24 MCP tools (TOOLS.md, generated and test-checked).

Boundaries

- Calendar fiscal year, straight-line AfA only.
- No direct booking: buchcraft receives drafts that a human approves.
- Not tax advice; categories and lives are defaults to check with the Steuerberater.

Gaps (honest)

- No egui window.
- Degressive AfA, Luxustangente (car cost cap), IFB and non-calendar fiscal years not modelled.
- Disposal sends no event (residual value booked by hand).
- No direct import from rust-devices-mcp or winops inventory (CSV only).
- SVG labels only, no label-printer driver; synthetic data only.

Next

1. egui app: register table, asset detail with schedule chart, expiry board, label print.
2. Steuerberater review; degressive AfA as an option.
3. Real inventory import from the fleet (Goliath, robots, plugs).

Complete MCP Tool & CLI Verb Reference

Generated by `assetcraft tools --markdown`. Do not edit by hand: a test fails when this file is out of date. Every tool is also a CLI verb (`assetcraft --help`). Dotted names (`app.verb`) are accepted.

Tool	What it does	Required	Optional
`asset_status`	Counts, cost of active assets, licence seats, open deadlines, GWG limit.	-	-
`asset_add`	Register an asset. cost is the AfA base (net if VAT is deductible). GWG is automatic at or below the limit unless gwg=false. Life defaults from the category; cars need at least 8 years.	`tag`, `name`, `in_service`, `cost`	`category`, `gwg`, `holder`, `invoice_ref`, `life_years`, `location`, `notes`, `purchase_date`, `serial`, `supplier`, `warranty_until`
`asset_list`	List assets with book value at the start of this year.	-	`category`, `include_disposed`, `text`
`asset_show`	One asset with AfA schedule, postings to buchcraft, maintenance and deadlines.	`tag`	`as_of`
`asset_update`	Change fields; null clears serial/location/holder/warranty_until. cost/in_service/life_years/gwg are frozen once AfA was posted.	`tag`	`cost`, `gwg`, `holder`, `in_service`, `life_years`, `location`, `name`, `notes`, `serial`, `warranty_until`
`asset_dispose`	Retire an asset (sold, scrapped). Returns disposal-year AfA, book value and gain/loss. The asset becomes read-only.	`tag`, `date`	`note`, `proceeds`
`asset_afa_schedule`	Straight-line AfA plan in cents with Halbjahres-AfA and GWG.	`tag`	-
`asset_afa_year`	AfA for all assets in a year with totals per account. emit=true sends asset.depreciation_due to buchcraft (drafts there, approved by a human); each asset/year is sent once.	-	`emit`, `year`
`asset_maintenance_add`	Log a repair, service or inspection with cost.	`tag`, `kind`	`cost`, `date`, `note`
`asset_category_list`	Categories with default life, minimum life and EKR asset account.	-	-
`asset_category_set`	Create or change a category (life 1-50, minimum life, EKR account).	`name`, `life_years`, `account`	`min_years`
`asset_gwg_limit`	Show or set the GWG limit (default 1.000,00 since 2023).	-	`set`
`asset_license_add`	Register a software licence. Never pass the key itself: key_location says where it is stored.	`name`, `seats`	`annual_cost`, `key_location`, `notes`, `renews_on`, `vendor`

Tool	What it does	Required	Optional
`asset_license_list`	Licences with assigned seats.	-	-
`asset_license_assign`	Give a person a seat. Refused when full unless force=true.	`license`, `assignee`	`force`
`asset_license_unassign`	Free a seat.	`license`, `assignee`	-
`asset_license_seats`	Change the number of seats.	`license`, `seats`	-
`asset_license_audit`	Over-allocated and unused seats, renewals due within 60 days, cost per used seat.	-	`as_of`
`asset_deadline_add`	Track a certificate, domain, insurance, inspection or contract date, optionally tied to an asset.	`name`, `due_on`	`kind`, `notes`, `tag`
`asset_deadline_done`	Mark a deadline handled.	`id`	-
`asset_expiries`	Warranties, licence renewals and deadlines due within N days (default 60), overdue first.	-	`as_of`, `within_days`
`asset_label`	Write a printable SVG label with a QR code (ASSET:<tag>) for an asset.	`tag`, `path`	-
`asset_import_csv`	Import assets from CSV tag;name;category;in_service;cost[;serial;location;holder]. All-or-nothing; dry_run validates only.	`path`	`dry_run`
`asset_demo_seed`	Fill an EMPTY register with a labelled demo inventory, licences and deadlines.	-	-